

Solution Requirements

Date	2 October 2026
Team ID	SWTID-2026-6582
Project Name	PocketSmart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration & Login	Users can create accounts, log in and log out; passwords are hashed and JWT cookies maintain authenticated sessions.
FR-2	Home Interior Planner	Accept budget and home requirements and generate personalized interior recommendations.
FR-3	Party Planner	Accept budget, guest count, event details and selected services and generate an event plan.
FR-4	Jewelry Planner	Generate jewelry recommendations from text preferences and optional outfit image.
FR-5	Gemini Integration	Send structured prompts to Gemini and validate structured AI output.
FR-6	Budget Validation	Calculate item subtotals and final totals in Python and ensure total estimated spending does not exceed budget.
FR-7	Recommendation History	Authenticated users can save, view, reopen and delete recommendation records.
FR-8	Search Links	Generate relevant external search URLs for suggested products/services.

Step 2: Non-Functional Requirements (NFR)

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Responsive, clear interface for desktop and mobile with readable forms and result cards.
NFR-2	Security	API keys stay in environment variables; passwords use bcrypt; JWT is stored in HTTP-only SameSite cookie.

NFR-3	Reliability	Malformed AI output and API failures are handled without silently substituting fake static recommendations.
NFR-4	Performance	Use lightweight server-rendered pages and Gemini Flash Lite for practical response latency.
NFR-5	Availability	Core web routes remain available; AI failures return a clear retry/error state.
NFR-6	Scalability	Modular routers/services and ORM allow migration to stronger databases/cloud hosting in future.